

exp/fig_d1.pdf

EXP D1 — foreground exhaustion		[scope:generator]
What may any null result be scoped to? If every image Stage C can add is a re-render of a foreground already in the base pool, a null reads “targeting does not help once the foreground pool is exhausted”.		
SETUP	MEASUREMENT	
raw HKU-IS 4447 foregrounds → LAKE-RED --isReplace frozen, seed 0 → authors' pool 4447 → local renders 4447 held fixed: the same 4447 masks feed both pools (raw.gt ≡ local.msk)	bijection: does every render trace to a base foreground, and is every base foreground rendered? literalness: object-interior vs background mean diff , every object, both pools	
OUTCOME	bijection 4447/4447 · 0 renders outside the base pool · 0 base foregrounds unrendered object interior 5.60 (local) / 9.63 (authors') vs background 70.80 / 70.55 ratio 12.6 × / 7.3 × 0 of 8885 objects regenerated the foreground is copied, not generated Consequence: every null in this paper is scoped to an <i>exhausted</i> foreground pool, and is silent on whether new foregrounds would change it.	

exp/fig_b1.pdf

EXP B1 — the allocation signal		[scope:signal+data]
Does student–teacher ES-disagreement – the signal that would drive targeted allocation – predict the error type that matters for camouflaged object detection (structure), or only pixel-average error?		
SETUP	MEASUREMENT	
target 4040 no ground truth → ES(stu, tea) per image a=0.9 b=0.3 → k-means k=75 dinoL518 COD10K-test has ground truth → ES on the endpoint → not available at allocation time the pipeline has no target labels, so the real signal is the <i>target</i>-measured one (CLS.py:81–105)	per-cluster Spearman ρ of ES against each error type, over 50 clusters / 3428 target images: MAE · 1 − S_α · 1 − IoU and the same correlation computed from the <i>endpoint</i>-measured signal, to size how much that overstates it	
OUTCOME	ρ (MAE) +0.8754 on the endpoint → +0.6284 where it is available (drop +0.2470) the two agree only at ρ +0.5732 per cluster ratio $\rho(1 - S_\alpha) / \rho(\text{MAE})$: 0.4352 → 0.5463 — it crosses B1’s own declared 0.5 boundary, in the opposite direction the signal is materially weaker where the method can use it and the cluster structure it allocates over is weak: best silhouette 0.1600	

exp/fig_c1.pdf

EXP C1 — targeted vs random allocation

[scope:signal+data]

How different is targeted training data from random? And if it is different, is that difference attributable to the uncertainty signal – or to something the signal is merely correlated with?

SETUP

target_es
over k=75 clusters

→

softmax $T=\alpha \text{sd}(es)$
budget, rank, select

B=1000 R2 cutout

held fixed in every arm: the allocation *shape* (same softmax, same budget, same ranking, same greedy rule). Only what the rule is pointed *at* changes.

each control destroys one thing: SHUFFLED the signal's cluster assignment · CENTROID the signal entirely · DIRECTION the fitted axis · NULL everything

MEASUREMENT

held-out Cohen's d , mean over 20 cells:

TARGETED (the real signal)

+0.675

SHUFFLED signal destroyed

+0.668

CENTROID arbitrary cluster

+0.740

DIRECTION random axis

+0.627

NULL random vs random

-0.133

OUTCOME

paired increment, targeted *minus* control, over all 20 cells:
vs its own SHUFFLED control +0.0073 of a d , targeted wins 13/20 — a coin flip
vs an arbitrary CENTROID −0.0649, targeted wins 4/20 — it is *worse*

destroying the signal reproduces the separation
what targeting buys is **concentration**, not targeting

exp/fig_abc.pdf

EXP ABC — the A/B/C campaign		[scope:generator]
Does a concentrated, ES-targeted Stage-C budget train better than a random one of the same size? Decided by a rule frozen before run 1: $\Delta > 2\hat{\sigma}$ and sign consistent 3/3 seeds.		
SETUP	MEASUREMENT	
base pool 4447 pairs 2 archs × 3 seeds = 24 runs → A0 +0 A2 +1000 B +1000 C10 +1000 unpadded baseline unclean: +22.5% exposure authors’ originals the clean control random renders redrawn per seed ES-targeted renders “Ours”, $\alpha=1.0$ total.step pinned 253 / 127 in both rounds of every run — added data buys 0 gradient steps	SINet COD10K-test (primary), S_α arm means: A0 A2 B C10 $B \pm 2\hat{\sigma}$, to scale A0 0.7010 · A2 0.7077 · B 0.7134 · C10 0.7185 — every arm lies <i>inside</i> the bar, and the whole A0→C10 span (0.0175) is narrower than the bar itself the pre-registered claim $\Delta(C-B) = +0.005034$, sign 3/3, against a bar of 0.017933	
OUTCOME	$\Delta(C-B)$ is WITHIN NOISE in all four arch × endpoint cells, and the two architectures disagree on its sign (SINet +0.005034, SINet-v2 −0.000399). The bar $2\hat{\sigma} = 0.017933$ is <i>larger</i> than the entire reference improvement of 0.0142 this design set out to half-resolve. no effect resolvable at this sensitivity a null bounded by power — not a demonstration of absence	

exp/fig_t2.pdf

EXP T2 — the falsification arms		[scope: signal+data]																				
A/B/C could not separate concentration from targeting. With the allocation shape held exactly fixed, does the direction of the uncertainty signal change trained accuracy at all?																						
SETUP	target_es over k=75 C10 identity ρ = +1 CSHUF permute ρ = −0.034 CINV reverse ρ = −1 identical allocation shape → 3 pools, 1000 images each the shape is not <i>approximately</i> equal — all four allocation keys reproduce C1’s committed cell at exact 5-dp equality (alloc.entropy_norm 0.78644, tv.from.uniform 0.49253, and two more), so the arms differ in the signal’s <i>direction</i> and in nothing else — which A/B/C structurally could not do.	MEASUREMENT																				
		Δ as a multiple of $\hat{\sigma}$, for every gap × architecture × endpoint: <table><tr><th></th><th>SI/C</th><th>v2/C</th><th>SI/N</th><th>v2/N</th></tr><tr><td>CSHUF–C10</td><td>0.55</td><td>0.83</td><td>0.62</td><td>0.66</td></tr><tr><td>CINV–C10</td><td>0.31</td><td>0.72</td><td>1.20</td><td>1.40</td></tr><tr><td>CINV–CSHUF</td><td>0.86</td><td>0.11</td><td>1.82</td><td>0.74</td></tr></table> every cell is below its bar of $2\hat{\sigma}$; the largest gap anywhere is 0.005007 (1.82 $\hat{\sigma}$)		SI/C	v2/C	SI/N	v2/N	CSHUF–C10	0.55	0.83	0.62	0.66	CINV–C10	0.31	0.72	1.20	1.40	CINV–CSHUF	0.86	0.11	1.82	0.74
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OUTCOME	12 of 12 cells WITHIN NOISE, at a bar $3.24\times$ tighter than A/B/C’s ($2\hat{\sigma} = 0.005533$ vs 0.017933 on SINet). Reported though not significant: C10 — the real signal — has the highest arm mean in none of the four cells, and CINV, the maximally anti-targeted arm, has it in three.	real, destroyed and reversed targeting are indistinguishable the signal’s direction carries no accuracy-relevant information at this sensitivity																				

exp/fig_t2c.pdf

EXP T2C — is the gap specific to ES?

[scope: signal+data]

target 4040
unlabelled

ES disagreement

predictive entropy

ensemble variance

one *shared*
boundary band

from the S2C teacher, so “boundary”
names the same region in every row

the pre-registered criterion, frozen before any T2-C number existed: each signal must keep $\rho(\text{MAE}) > \rho(1 - S_\alpha) > \rho(1 - \text{IoU})$ over B1’s committed 50-cluster partition.

instrument check: the ES / SINet / whole cell reproduces B1’s committed $\rho(\text{MAE})$ **+0.6284** *bitwise*, before any new number is read.

MEASUREMENT

ordering held? (seed 0)

WHOLE-IMAGE

BOUNDARY

SINetv2SINetv2

ES

entropy

ens. A0

ens. CSHUF

PASSPASSFAILFAIL

PASSPASSFAILPASS

PASSPASSPASSFAIL

PASSPASSFAILFAIL

whole-image 8/8, 7 of them at 10/10 *k*-means seeds. Boundary 2/8, and *four* rows go negative.

OUTCOME

The ordering is **not** a property of ES: it holds for predictive entropy and ensemble disagreement too. But the *strong* reading fails. $\rho(1 - S_\alpha)$ reaches **+0.5628** on SINet-v2, so “uncertainty carries no localisation information” is **false as stated**. The claim is **ordinal and comparative only**.

the gap generalises; the strong reading does not correlational, one endpoint — it cannot become “uncertainty guidance fails in training”

exp/fig_d2.pdf

EXP D2 + D2R + D2_NC4K — benchmark contamination

[scope: method-independent]

Is the CHAMELEON evaluation set re-encoded copies of the data these models train on? And does the detector find contamination everywhere it looks, or can it return a clean answer?

(a) THE DETECTOR — exact hashing is the wrong instrument

byte hash
0 found

pixel hash
0 found

same-dimension
near-duplicate

41 / 76 (53.9%)

DESC=32 RMS<=14 TOL=6.0

exact hashing calls CHAMELEON *perfectly clean* while 53.9% of it is re-encoded training data

(b) 41 IS A PROPERTY OF THE DATA, NOT OF THE CUTOFF

tolerance sweep saturates:

1126374041

1.02.03.05.06.0

tolerance

a *second, independent* instrument agrees on the same integer: the sorted nearest-distance list has **41 below 5.51** and the next at **40.58** — a **7.36×** jump, and the author-sourced copy is byte-identical (76/76).

(c) THE NEGATIVE CONTROL — the same detector, imported unchanged

NC4K 4121
vs COD10K-test

0 / 4121

and the honesty bound: 3039 of 4121 NC4K images (73.7%) have no same-dimension candidate at all, so they are **unchecked, not clean**. Every rate here is a *lower bound*.

OUTCOME

CHAMELEON is **not an independent endpoint** for COD10K-trained models — and that is the whole claim. D2R measured for a score-inflation figure and **found none** (all eight percentiles in 0.448–0.559, and the sign flips with the mask release), so none is reported.

independence, not inflation
a protocol violation does not need a score advantage to be disqualifying

exp/fig_a3.pdf

EXP A3 — synthetic vs real

[scope: generator] [scope: method-independent]

How far do LAKE-RED’s renders sit from the real target distribution – and does the metric that was used to say so survive an honest control?

(a) MANIFOLD COVERAGE — *k*-NN recall vs the real target set, dinoL518, *k*==5

ceiling (real vs real)

0.9431

NC4K – a different real set

0.8995

raw HKU-IS: its OWN input pool

0.7097

authors’ synthetic pool

0.4834

synthetic covers **0.4834** where **its own input pool covers 0.7097**
a paired within-content loss of **–31.9%**; the render moves the pool *away* from the target

the comparison that matters is against LAKE-RED’s own input, not against the ceiling — it is the only one that holds content fixed

(b) BUT THE METRIC THAT USED TO CARRY THIS IS NEAR-VACUOUS — clipL224

real vs LAKE-RED (the headline)

0.9995

real vs THE SAME images, JPEG-30

0.9928

real vs real, random halves (true null)

0.4717

re-encoding the *identical* images separates them from themselves at **0.9928**
within **0.0067** of the headline, on content that is identical by construction

and the estimator manufactures effects: on a *true null* the in-sample *d* is **+0.6867** while held-out it is **+0.0318**

OUTCOME

The **finding survives**; the metric that used to carry it does **not**. The coverage loss is real and passes 3/3 embedders. The AUC framing is withdrawn — a near-1.0 separation is not evidence that two image sets differ distributionally.

a **distributional characterisation, not a training-utility bound**
“far, therefore useless” is a non-sequitur